MODULE 6

NULL HYPOTHESIS SIGNIFICANCE TESTING

NULL HYPOTHESIS SIGNIFICANCE TESTING

- Null hypothesis significance testing (NHST) is a procedure that uses a sample of data to determine whether an effect or association might be present at the population level
- A cornerstone of psychological research for over 100 years
- Provides a formal recipe to evaluate evidence from data, and the same steps apply to virtually any research question

SIGNIFICANCE TESTING STEPS

1. Specify hypotheses
2. Define standard of evidence
3. Design study and collect data
4. Compare data to null hypothesis
5. Evaluate hypotheses and draw conclusion

OUTLINE

1. Quick review
2. Overview of NHST
3. Significance testing steps
4. Study questions

OUTLINE

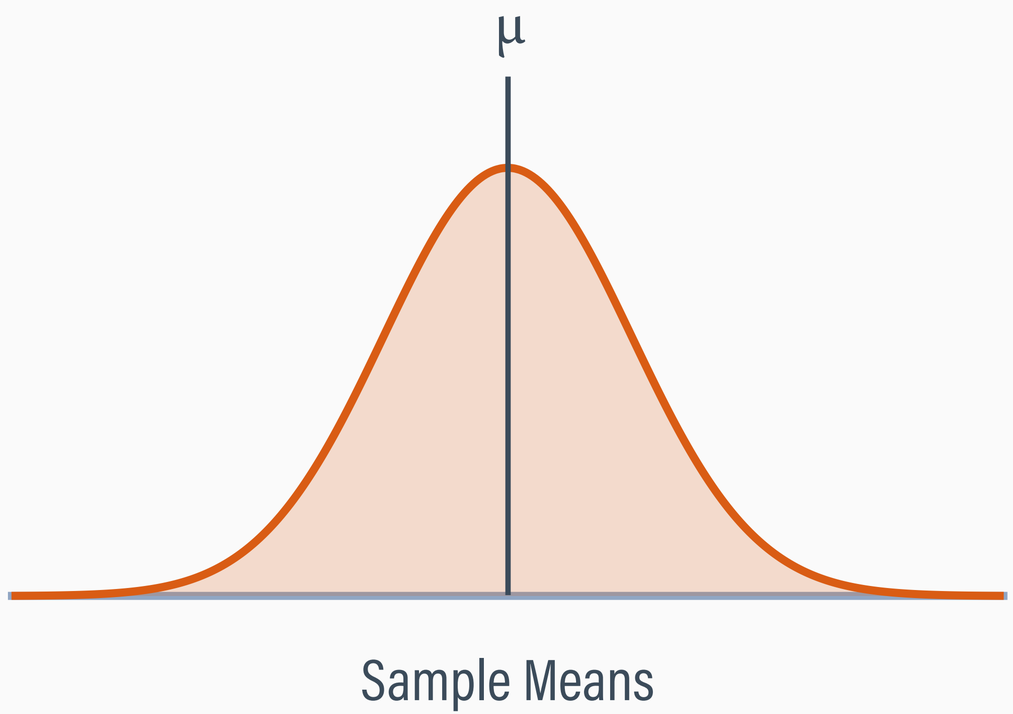
1. Quick review
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QUICK REVIEW: SAMPLING ERROR

- The frequentist paradigm imagines a single population that spawns many hypothetical random samples of data (one parameter, many hypothetical estimates)
- The amount by which an estimate differs from the true population statistic is called sampling error
- Every hypothetical sample has a different amount of error

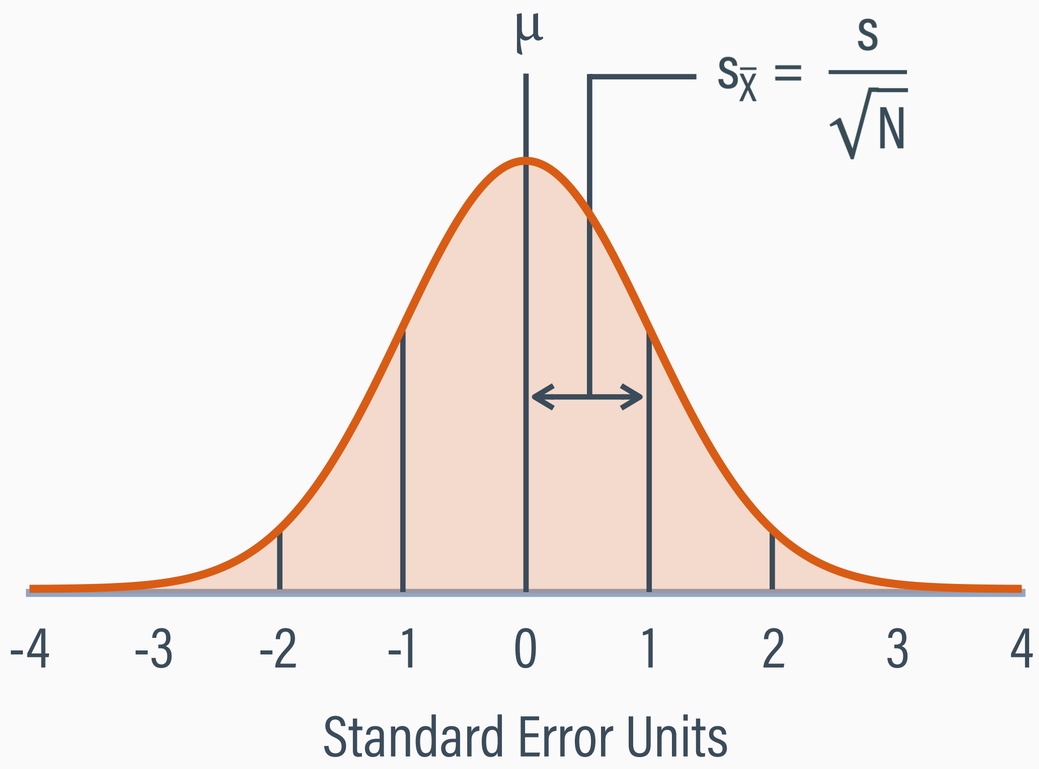
QUICK REVIEW: SAMPLING DISTRIBUTION

- The distribution of the estimates from many hypothetical samples is a sampling distribution
- With a large enough N, sample means follow a normal curve centered at the true mean
- Most estimates have small sampling errors, but a few have larger errors



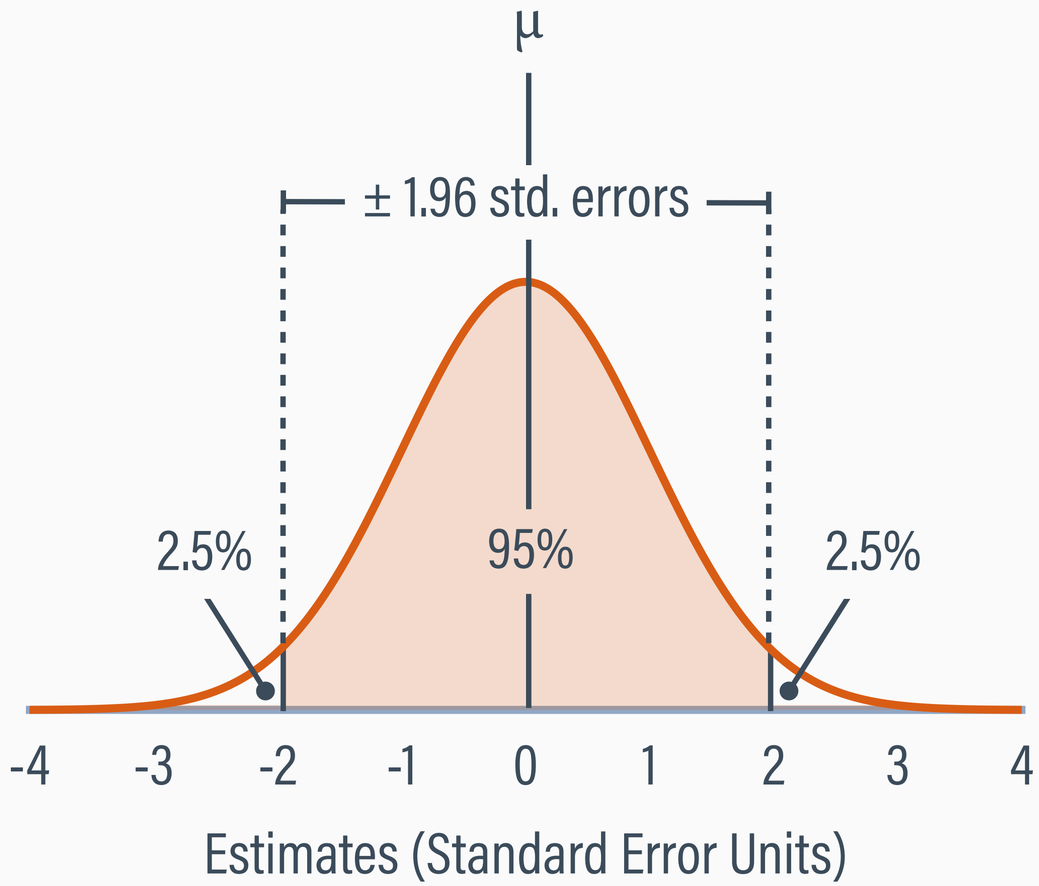
QUICK REVIEW: STANDARD ERROR

- The standard error is the average distance from a sample mean and the true mean
- $s_{\bar{X}}$ = standard deviation of the sample means
- The standard error is the average or expected amount of sampling error across many hypothetical samples



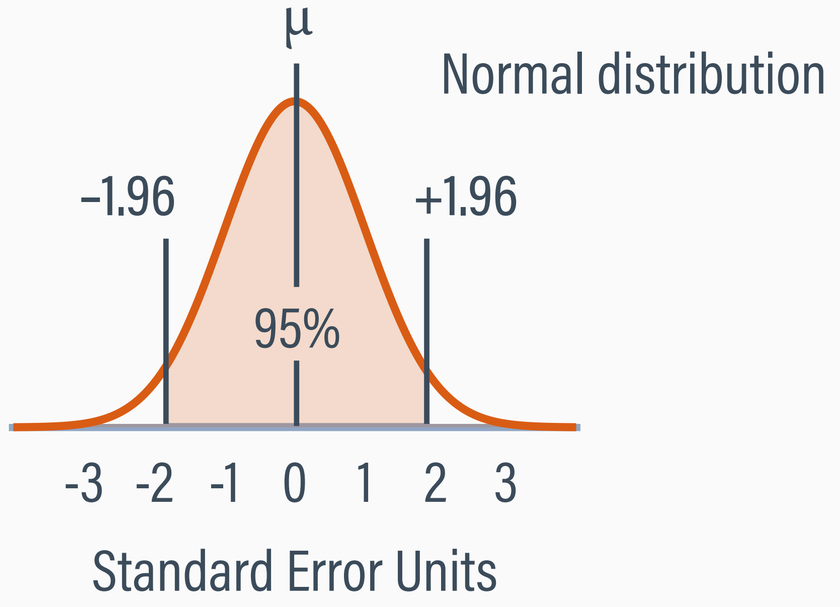
QUICK REVIEW: NORMAL CURVE RULE

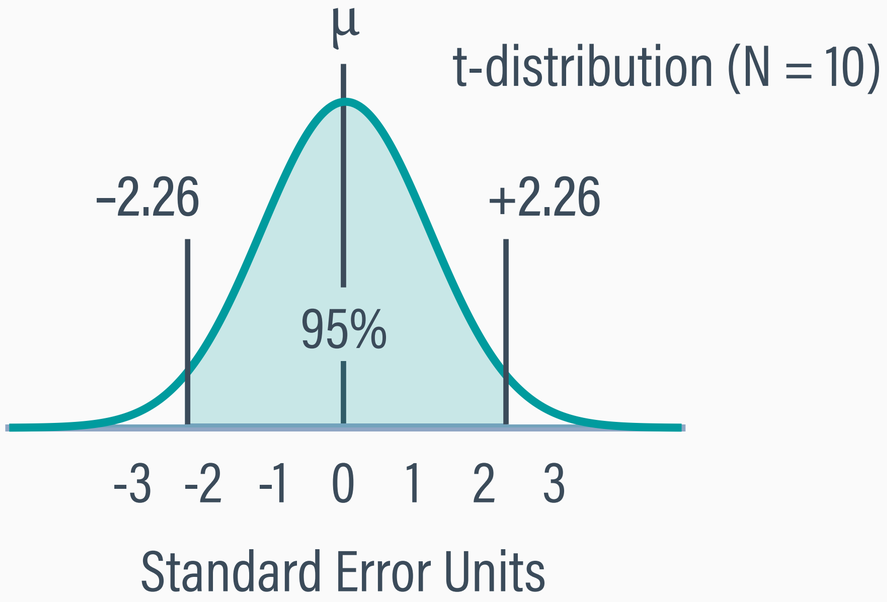
- The standard error is the standard deviation of many hypothetical sample means
- We can apply normal curve rules of thumb
- 95% of the means from large samples are within ± 1.96 standard errors of the true mean



QUICK REVIEW: T-DISTRIBUTION

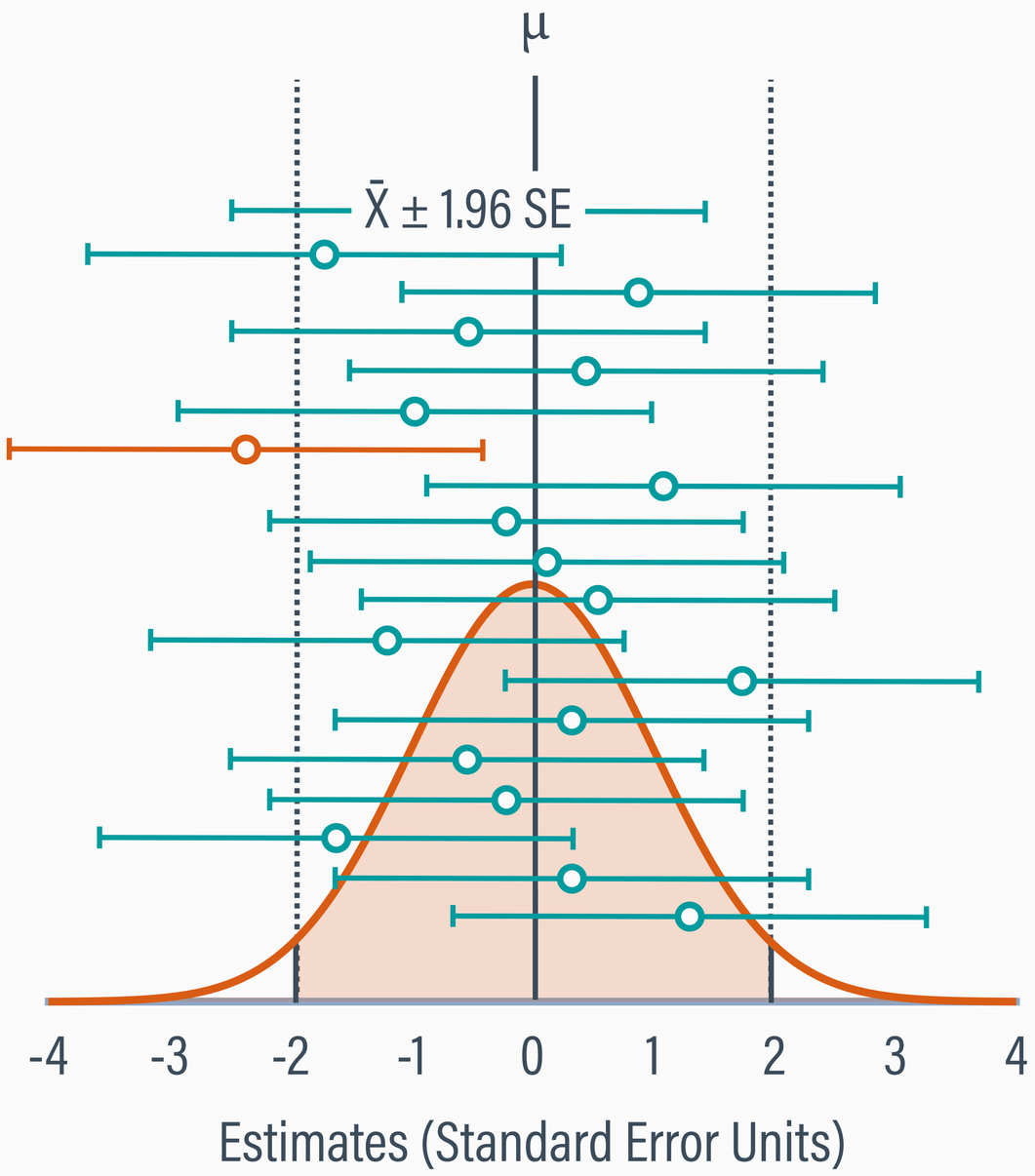
- When using small samples, the normal curve is an inaccurate description of sampling error
- The t-distribution is a series of bell-shaped curves that stretch out (become more variable) as the N decreases
- Small samples are more likely to produce outlier estimates, and “stretching” the curve honors that





QUICK REVIEW: CONFIDENCE INTERVALS

- Sliding the 95% range around $\bar{X}$ defines the most extreme population means that could have reasonably produced our data
- “95% confident” = across many hypothetical samples, 95% of the intervals should include the true population mean



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UVEAL MELANOMA and DEPRESSION

Uveal melanoma, a rare eye cancer, presents potential vision loss and life threat. This prospective, longitudinal study interrogated the predictive utility of visual impairment, as moderated by optimism/pessimism, on depressive symptoms in 299 adults undergoing diagnostic evaluation.



Annette Stanton

James MacDonald

MacDonald, J.J., Jorge-Miller, A., Enders, C.K., McCannel, T., Beran, T., & Stanton, A.L. (2021). Perceived and objective visual impairment predicting depressive symptoms across one year in uveal melanoma diagnostic biopsy: Optimism and pessimism as moderators. *Health Psychology*, *40*, 408-417.

KEY VARIABLE



Depression

The CES-D is a 20-item inventory that asks people to rate how often they experience depressive symptoms such as restless sleep, poor appetite, and feeling lonely.

RESEARCH QUESTION

- Studies typically attempt to answer a handful of research questions involving associations between key variables
- Do adults undergoing diagnostic evaluation for a possible eye cancer experience depressive symptoms at the level typical of cancer patients in treatment?
- The literature suggests that CES-D scores ~ 13 are normative for cancer patients

NULL HYPOTHESIS

- The significance testing process starts with a null hypothesis that specifies a baseline value for a population parameter
- The null hypothesis usually corresponds to no change, no effect, etc.
- The literature suggests that cancer patients receiving treatment have a mean CES-D score of about 13
- The null says that adults evaluated for a possible eye cancer are no different: their population mean is also $\mu = 13$

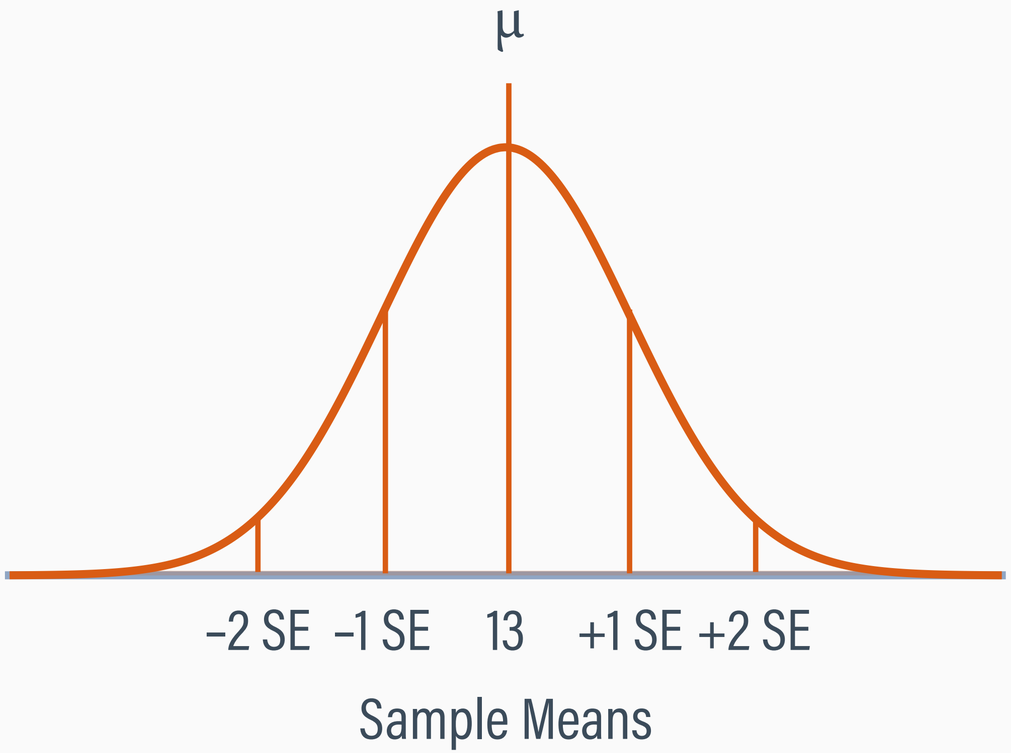
DEMONSTRATING AN EFFECT

- To demonstrate an effect (e.g., people evaluated for a possible eye cancer are differ from other patients in treatment), the data must disagree with the null

- Refuting the null that $\mu = 13$ requires a sample $\bar{X}$ that is much higher or much lower than 13

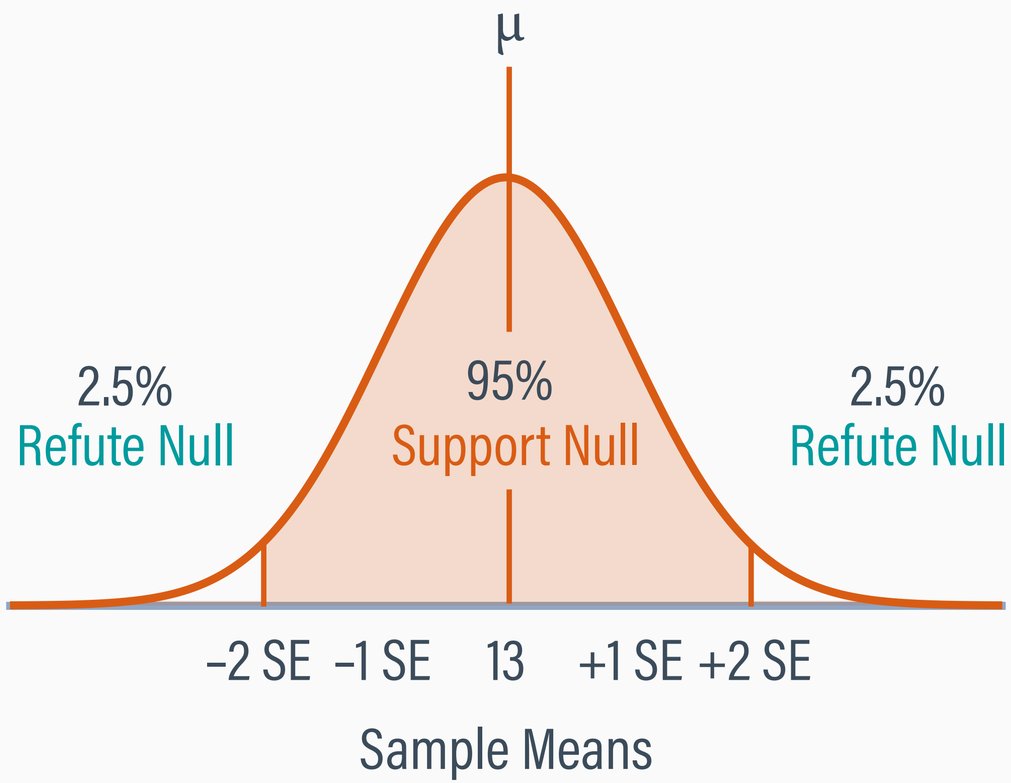
SAMPLING DISTRIBUTION IF NULL IS TRUE

- With the standard error, we can construct the sampling distribution around the null value of μ
- The sampling distribution shows the means from many hypothetical samples of N = 299 drawn out of the null population with $\mu = 13$
- From the sampling distribution, we can determine how close $\bar{X}$ is to the null μ



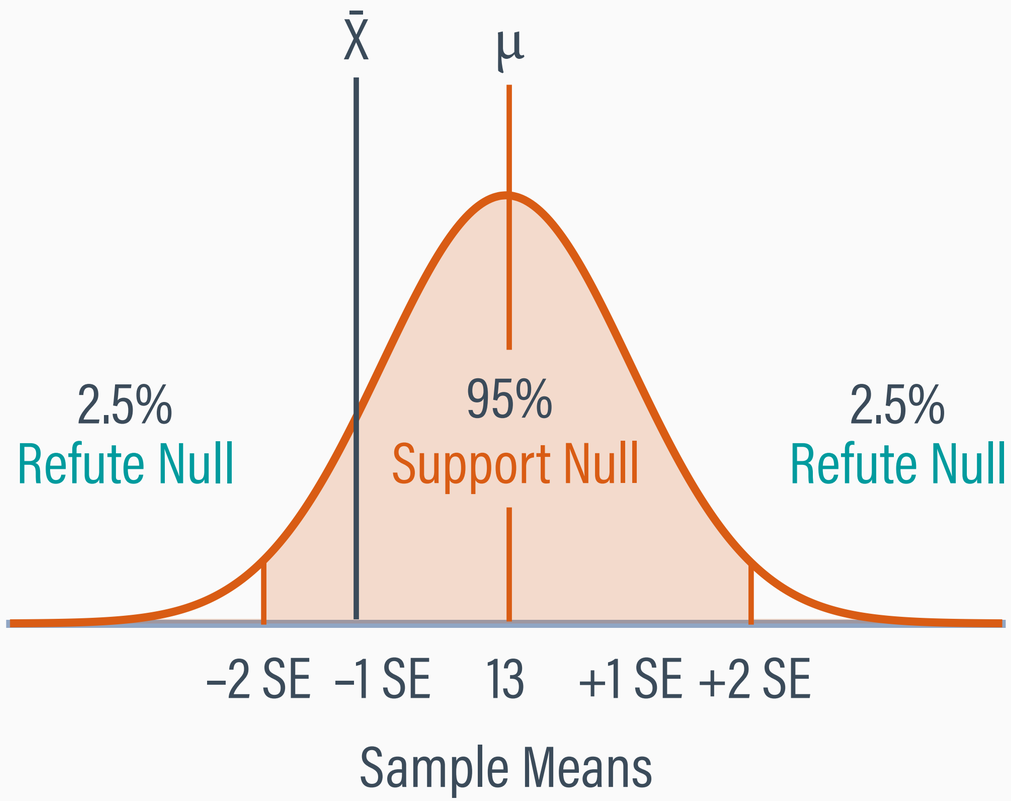
EVALUATING THE NULL

- By convention, we refute the null if the sample $\bar{X}$ falls outside the middle 95% of the sampling distribution
- Such as sample has less than a 5% chance of originating from the null population
- Any $\bar{X}$ within the middle 95% of the sampling distribution lends support to the null



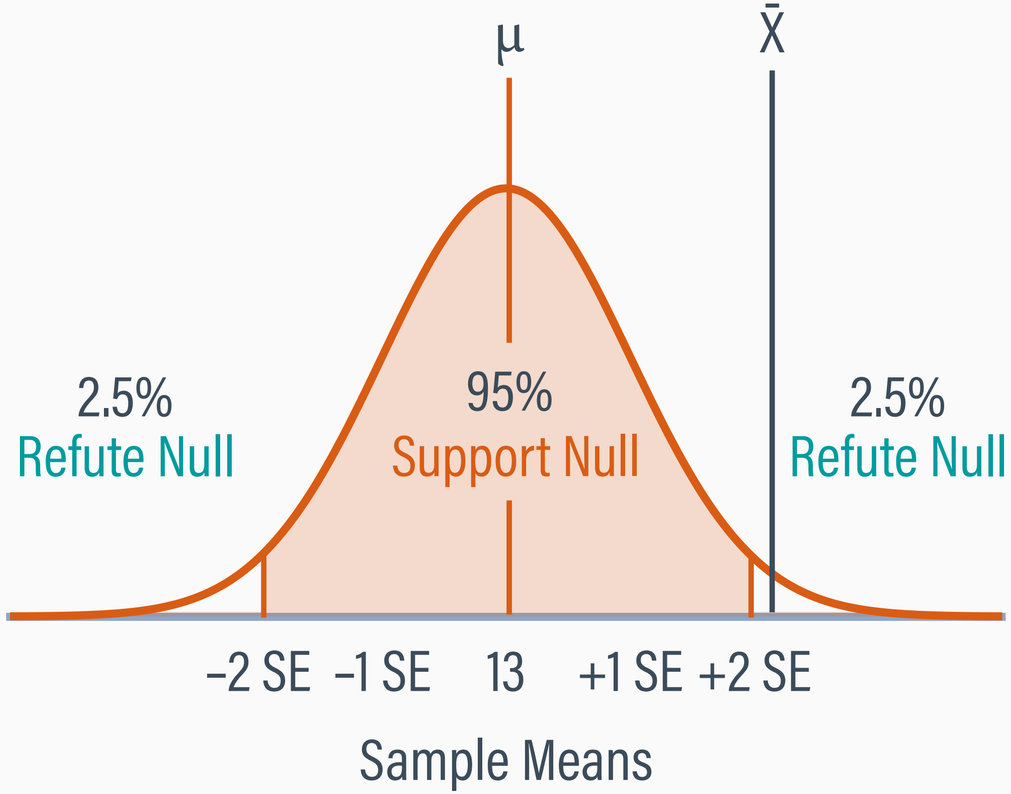
SUPPORT THE NULL

- Any sample within the 95% region around the null value of could reasonably originate from the normative population
- The $\bar{X}$ in the graph is consistent with other sample means from the null population
- The data lend support to the null



REFUTE THE NULL

- Any sample within the 95% region around the null value of could reasonably originate from the normative population
- The $\bar{X}$ in the graph is different from other sample means from the null population
- The data contain evidence against the null



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NULL HYPOTHESIS

- In the population, people treated for cancer have a mean equal to $\mu = 13$

$$H_0: \mu = 13$$

- The null is a “no effect” hypothesis in the sense that it represents a normative level for other cancer patients

ALTERNATE HYPOTHESIS

- The alternate hypothesis reflects the researcher's expectations about an effect
- A one-tailed (directional) alternate hypothesis predicts an effect in just one direction (e.g., depression is higher than the norm)
- A two-tailed alternate predicts an effect in either direction (e.g., depression is higher or lower than the norm)
- Two-tailed tests predominate psychology applications

TWO POSSIBLE ALTERNATIVE HYPOTHESES

- One-tailed alternate: Adults evaluated for a possible eye cancer are more depressed than the norm for cancer patients in treatment

$$H_A: \mu > 13$$

- Two-tailed alternate: Adults evaluated for a possible eye cancer could be more or less depressed than other cancer patients

$$H_A: \mu \neq 13 \ (\mu > 13 \text{ or } \mu < 13)$$

ROLE OF THE ALTERNATE HYPOTHESIS

- We do not directly test the alternative hypothesis, which represents the researcher's belief
- We provide support for the alternative by rejecting the null
- Whether the alternative is one-tailed or two-tailed is very important, as this determines what kind of evidence would lead us to reject the null

SIGNIFICANCE TESTING STEPS

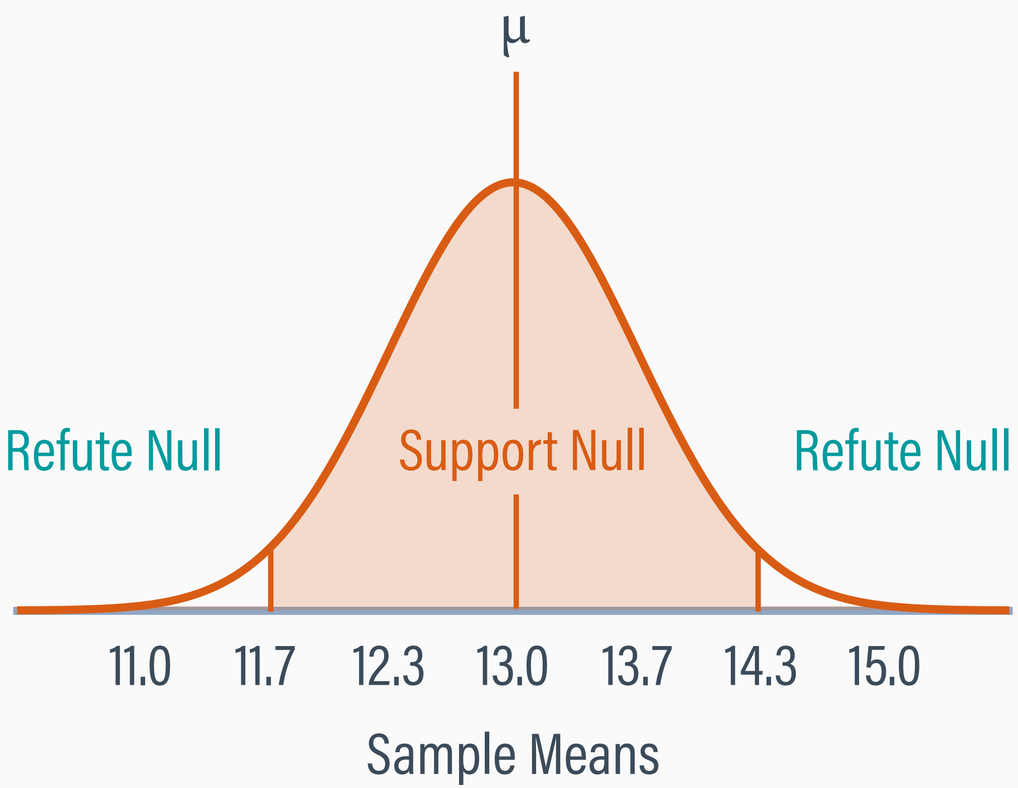
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STANDARD OF EVIDENCE

- The data are the evidence that we use to conclude whether the null is plausible (“innocent”) or implausible (“guilty”)
- If the sample mean from our data is very different from the null mean, then we conclude that the null hypothesis is implausible
- How big a difference do we need to observe to refute the null?

EVALUATING THE NULL

- Any $\bar{X}$ near the middle of the sampling distribution lends support to the null
- Such as sample has a high probability of originating from the null population
- We refute the null if the sample $\bar{X}$ falls far from μ
- Such as sample has a low probability of originating from the null population



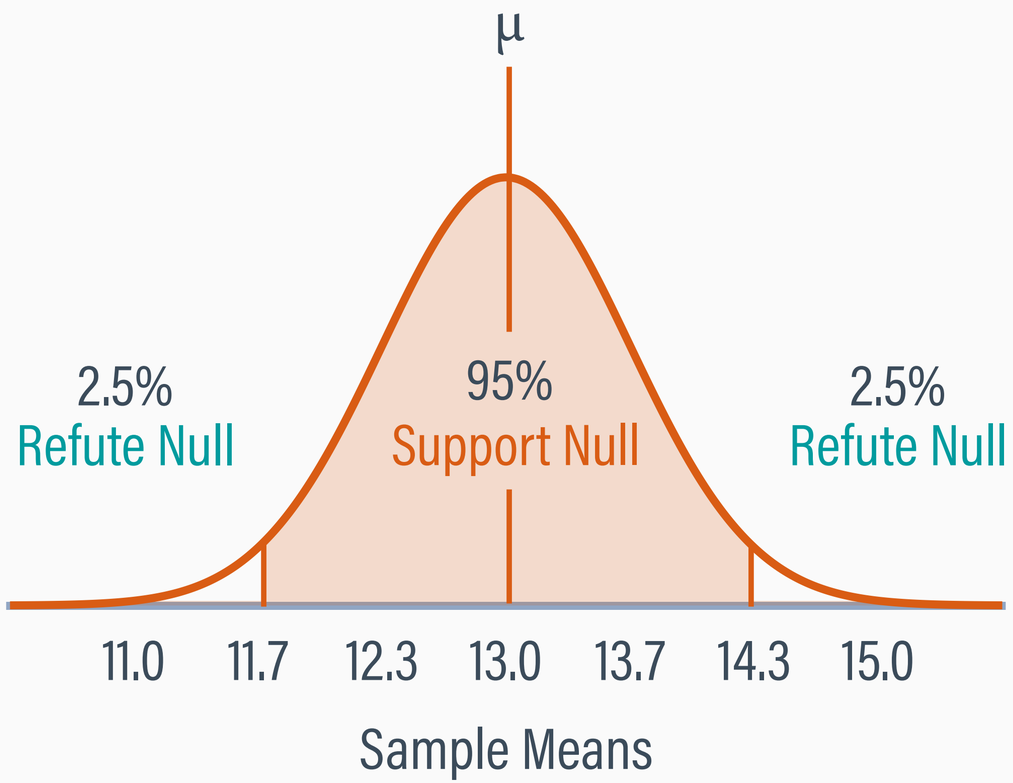
5% SIGNIFICANCE CRITERION



- Evidence against the null is presented as a probability
- R.A. Fisher — an influential biologist and statistician from the early 1900s — recommended 5% ($p < .05$) as a criterion
- If a sample mean has a 5% or less probability of originating from the null population, then we conclude H_0 is implausible

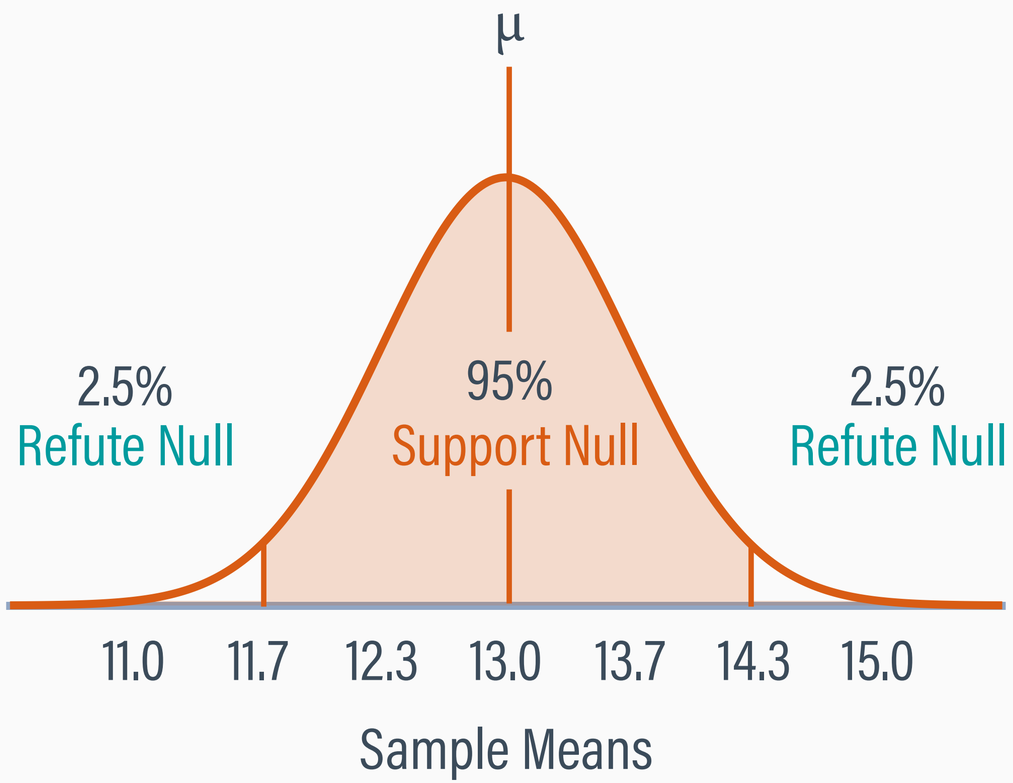
SAMPLING DISTRIBUTION

- By convention, we refute the null if the sample $\bar{X}$ falls outside the middle 95% of the sampling distribution
- Such as sample has less than a 5% chance of originating from the null population
- That is, we refute the null if the sample $\bar{X}$ is an outlier (rare) from the null population ($p < .05$)



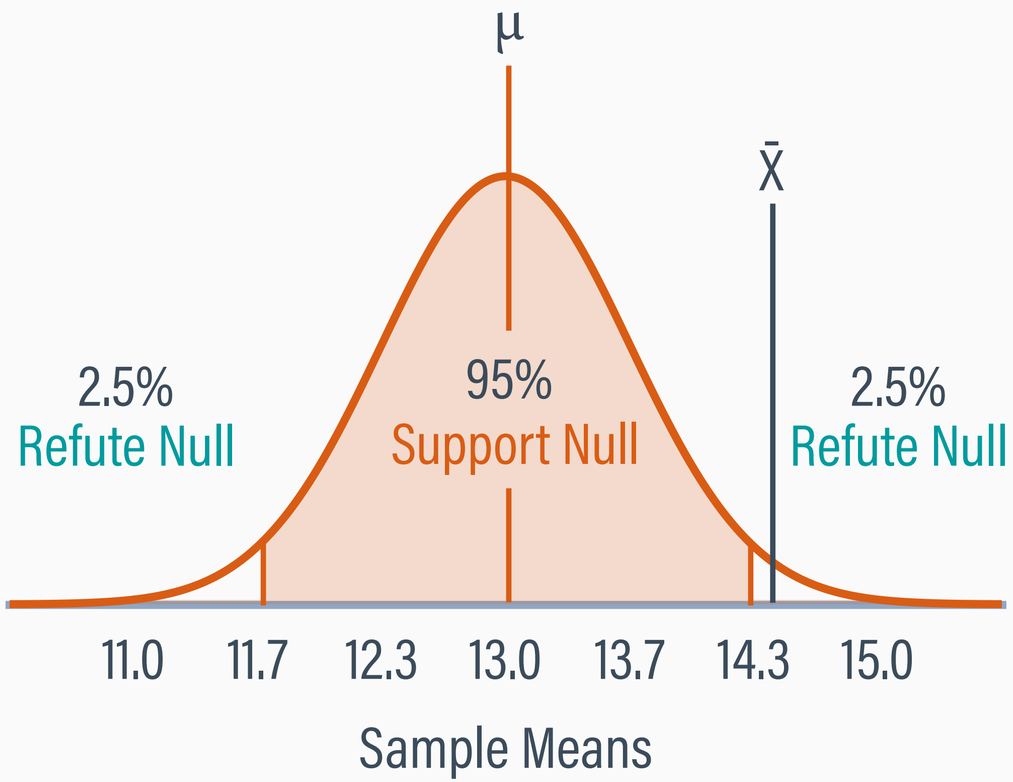
TWO-TAILED ALTERNATE HYPOTHESES

- A two-tailed alternate hypothesis predicts an effect in either direction
- e.g., People evaluated for a possible eye cancer could be above or below the treatment benchmark
- The 5% rejection region (alpha level) is split in half to allow for the possibility that either an increase or a decrease provides evidence against H_0

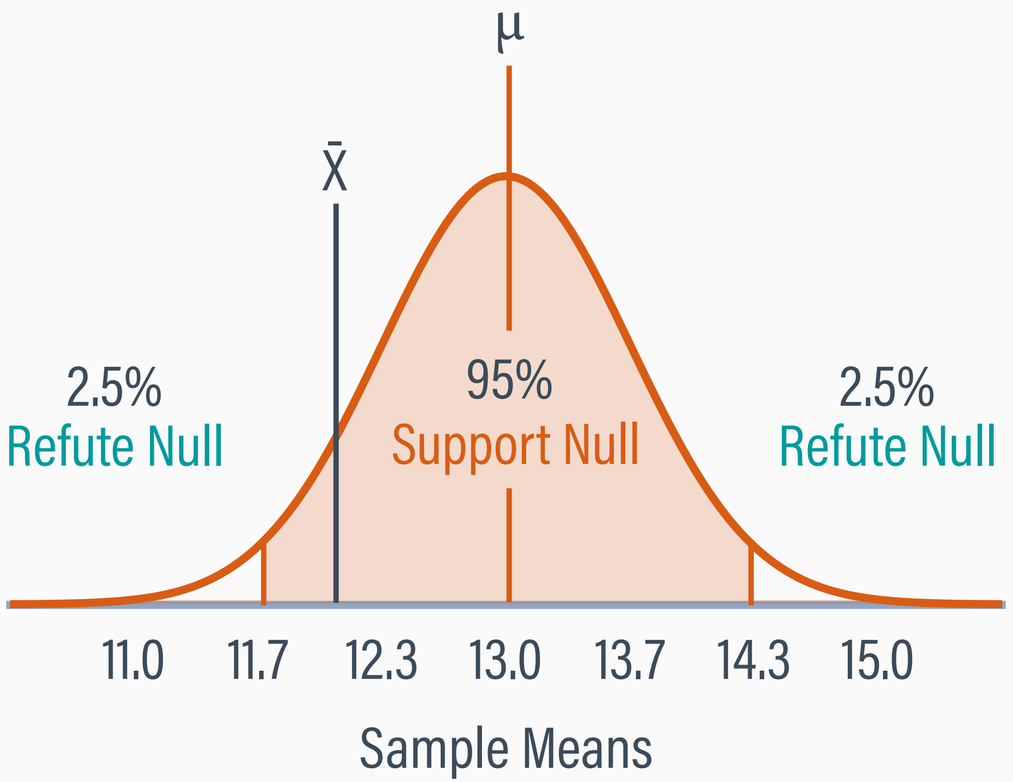


TWO POSSIBLE DECISIONS

Refute the null: $\bar{X}$ is unlikely from a population with $\mu = 13$ ($p < .05$)

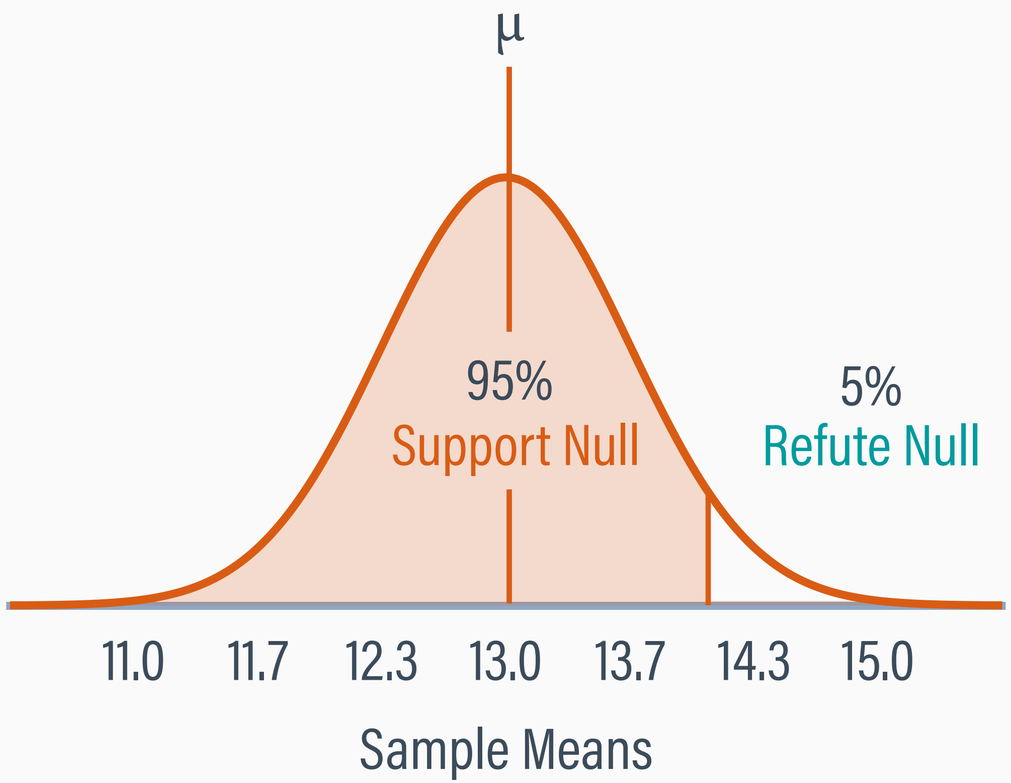


Support the null: $\bar{X}$ is likely from a population with $\mu = 13$ ($p > .05$)



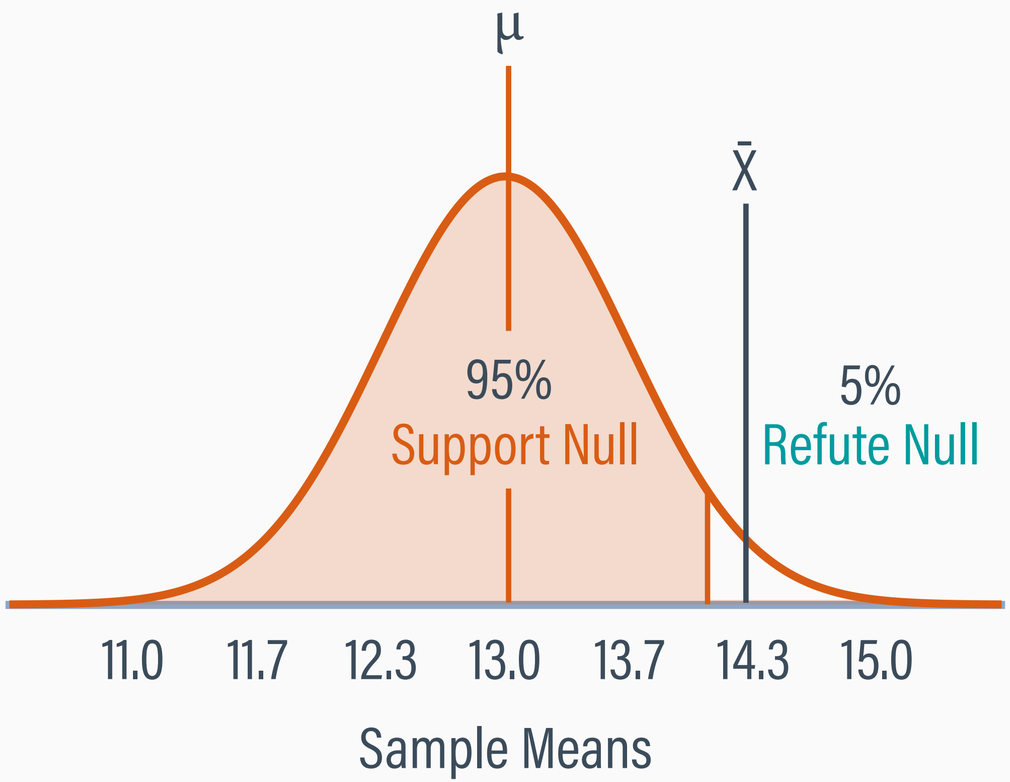
ONE-TAILED ALTERNATE HYPOTHESES

- A one-tailed alternate hypothesis predicts an effect in just one direction
- e.g., People evaluated for a possible eye cancer could only be above the treatment benchmark
- The 5% rejection region (alpha level) is assigned to one tail to reflect that only a very high $\bar{X}$ can provide evidence against H_0

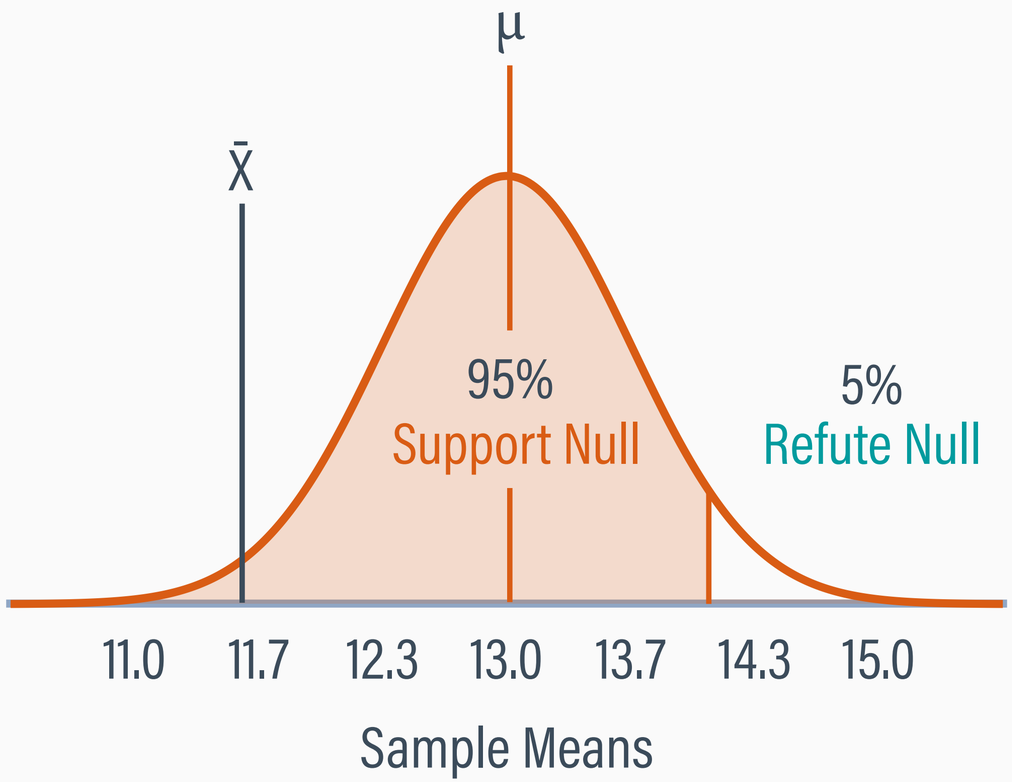


TWO POSSIBLE DECISIONS

Refute the null: $\bar{X}$ is unlikely from a population with $\mu = 13$ ($p < .05$)



Support the null: $\bar{X}$ is likely from a population with $\mu = 13$ ($p > .05$)



FALSE POSITIVES (TYPE I ERRORS)

- The 5% criterion defines a region of the distribution that contains outlier samples that are unlikely *but not impossible*
- When $\bar{X}$ falls in the rejection region (evidence against the null), there is still a 5% chance it came from the null population
- A significant result means we risk a false positive—incorrectly rejecting the null when it is actually true (called a Type I error)
- Adopting a 5% criterion fixes the Type I error risk to 5%

SIGNIFICANCE TESTING STEPS

1. Specify hypotheses
2. Define standard of evidence
3. Design study and collect data
4. Compare data to null hypothesis
5. Evaluate hypotheses and draw conclusion

ANALYSIS SUMMARY

- N = 299 participants evaluated for eye cancer
- The sample mean is 14.85
- The standard error $s_{\bar{X}} = 0.66$ indicates that the expected (average) sampling error across many samples of N = 299 is about 0.66 CES-D points

N	$\bar{X}$	SD	SE
299	14.85	11.44	0.66

$$s_{\bar{X}} = \frac{s}{\sqrt{N}} = \frac{11.44}{\sqrt{299}} = 0.66$$



Consider the sampling distribution of sample means from a null population with $\mu = 13$. In small groups of two or three, use the mean and standard error to determine whether $\bar{X} = 14.85$ ($s_{\bar{X}} = 0.66$) provides evidence for or against the null hypothesis.

SIGNIFICANCE TESTING STEPS

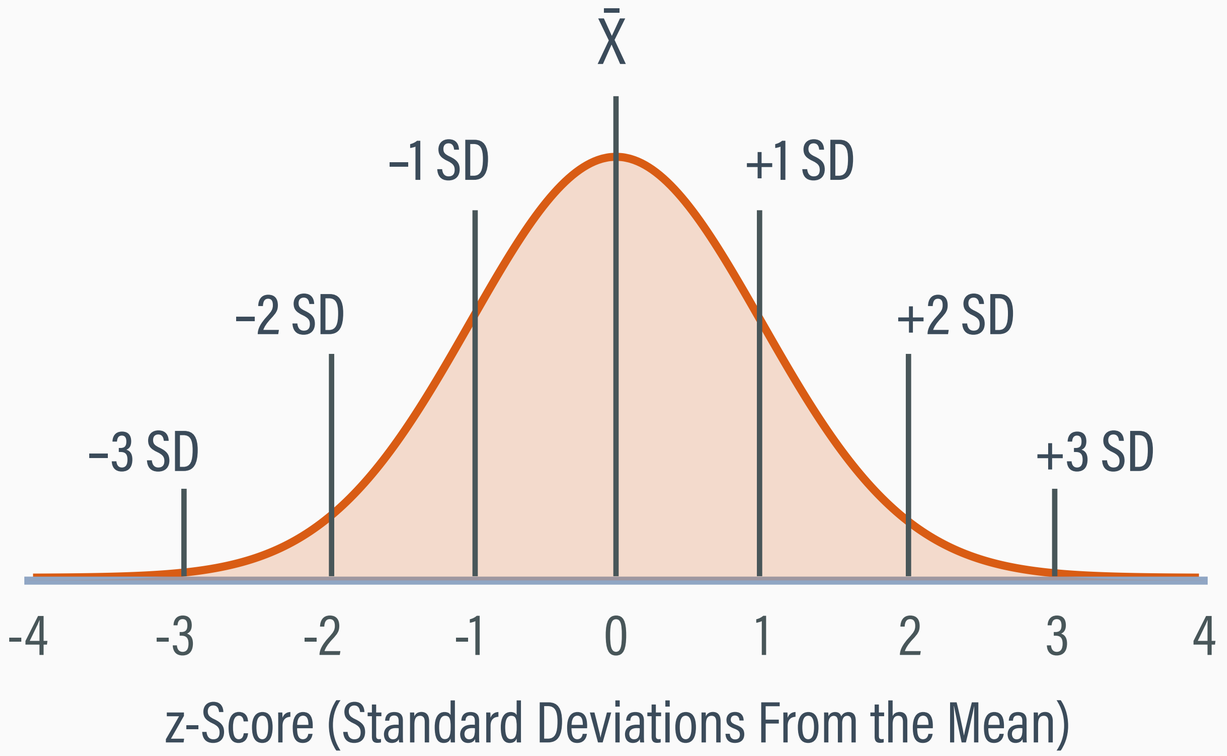
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COMPARING DATA TO THE NULL

- Two ways to determine whether the sample $\bar{X}$ is consistent (or inconsistent) with the null population mean
- The t-statistic gives a standardized distance between the sample mean and the null hypothesis mean (like a z-score)
- A p-value tells us how likely it is that hypothetical samples like our data would originate from the null population

z-SCORES REVISITED

- The z-score scale is a common standardized metric that expresses scores as standard deviation units from the mean



z-SCORE FORMULA

- The numerator captures the raw score's distance (deviation) from the sample mean $\bar{X}$

$$z = \frac{X - \bar{X}}{s} = \frac{\text{distance from mean}}{\text{standard deviation}}$$

- Dividing by the standard deviation converts the distance to standard deviation units

t-STATISTIC

- The t-statistic quantifies the number of standard error units that separate the sample mean and null hypothesis population mean

$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{\text{distance from the null}}{\text{standard error (std. dev. of } \bar{X})}$$

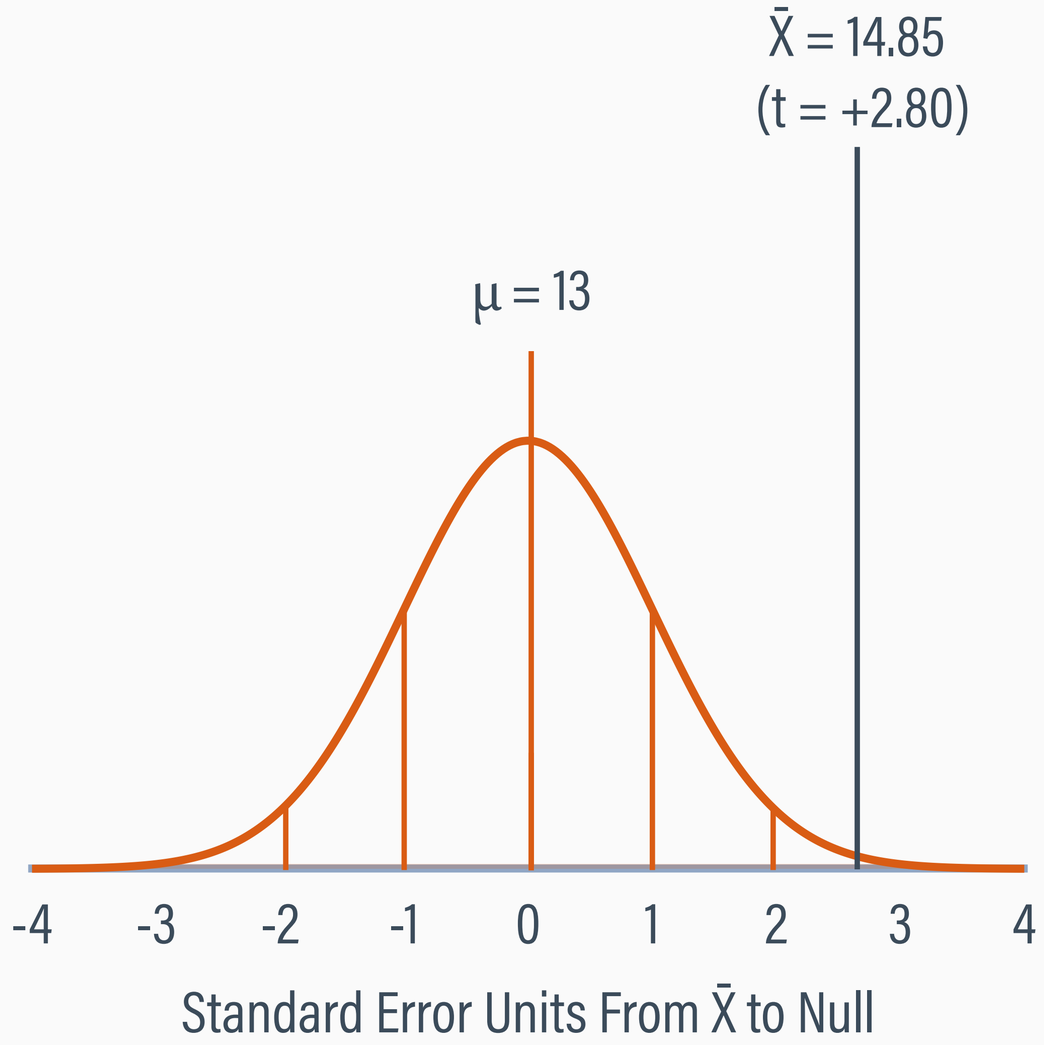
- The t-statistic is the same as a z-score (a standardized metric where distance is expressed in standard deviation units)

t-STATISTIC EXAMPLE

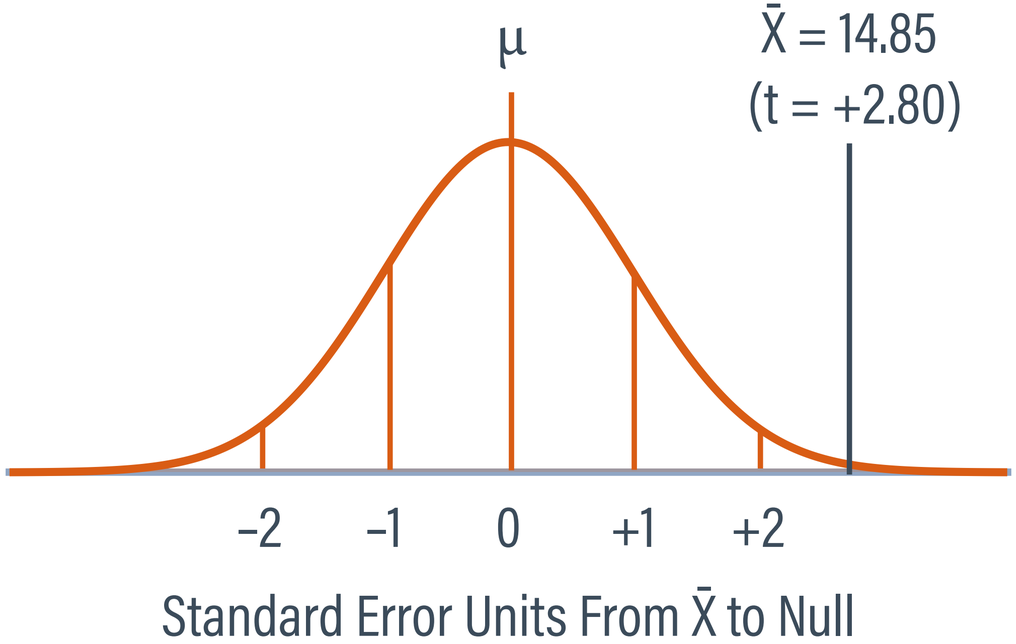
- The positive t-statistic conveys that the sample $\bar{X}$ is higher than the null ($\mu = 13$)

$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{14.85 - 13}{0.66} = 2.80$$

- The numeric value indicates that $\bar{X}$ is 2.80 standard error units from the null null







Consider the sampling distribution of sample means from a null population with $\mu = 13$. The sample mean and t-statistic are $\bar{X} = 14.85$ and $t = 2.80$. In small groups of two or three, apply the normal curve rule of thumb and decide whether the sample data provide evidence for or against the null hypothesis

FREQUENTIST PARADIGM REVISITED

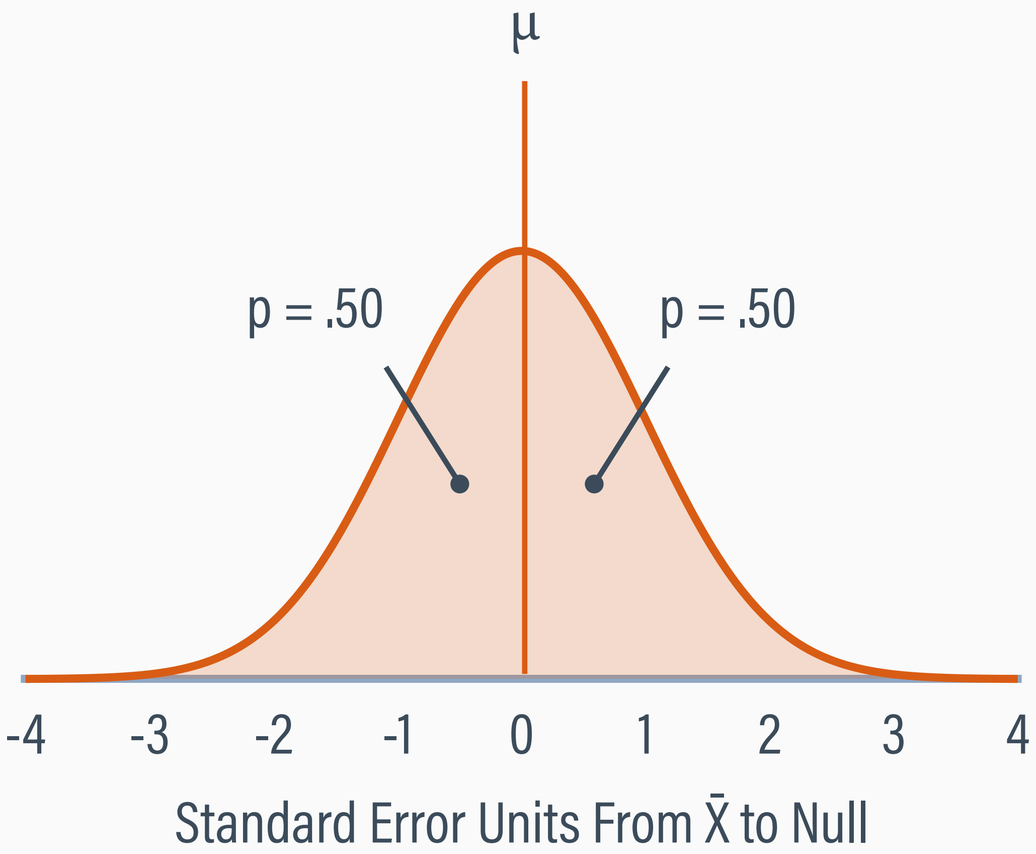
- The frequentist paradigm is defined by the idea that there is one population with unknown parameters (e.g., μ and σ)
- We imagine numerous hypothetical samples of size N from the population, each with its own unique estimates and 95% CIs
- The population-level statistics (parameters) are locked in at a single set of values, whereas the sample-level statistics (estimates, t-statistics) vary across different data sets

PROBABILITY VALUES (P-VALUES)

- A p-value is defined as proportion of hypothetical samples that have a t-statistic at least as large as the sample data
- Assuming the null is true, how likely is it to draw a sample with an effect at least as large as the one from our data?
- Visually, probability is an area under the curve, obtained by applying calculus integrals to the t-distribution function

TOTAL AREA UNDER THE CURVE

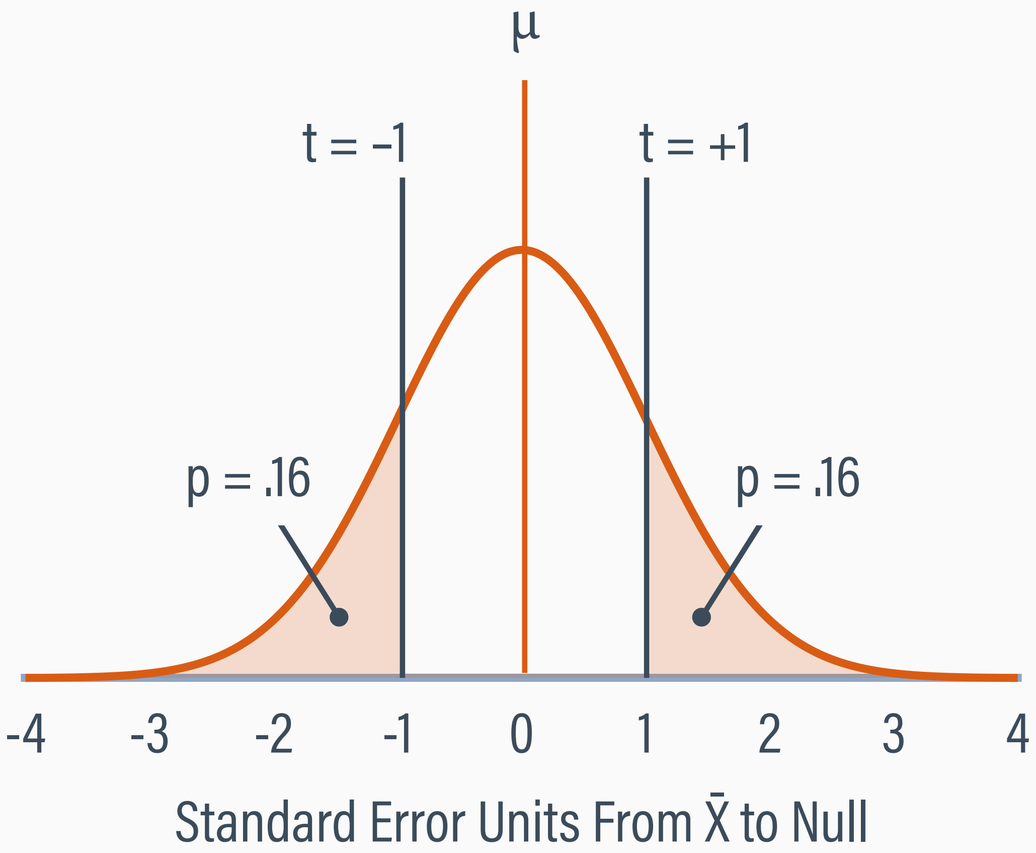
- The total probability (area under the sampling distribution) in both directions is 1.0
- 50% of all hypothetical samples are above the null population mean and 50% are below



P-VALUE AT ± 1 STANDARD ERRORS

- 16% of all hypothetical samples are +1 or more standard error units above the null mean, and 16% are –1 or more standard errors below

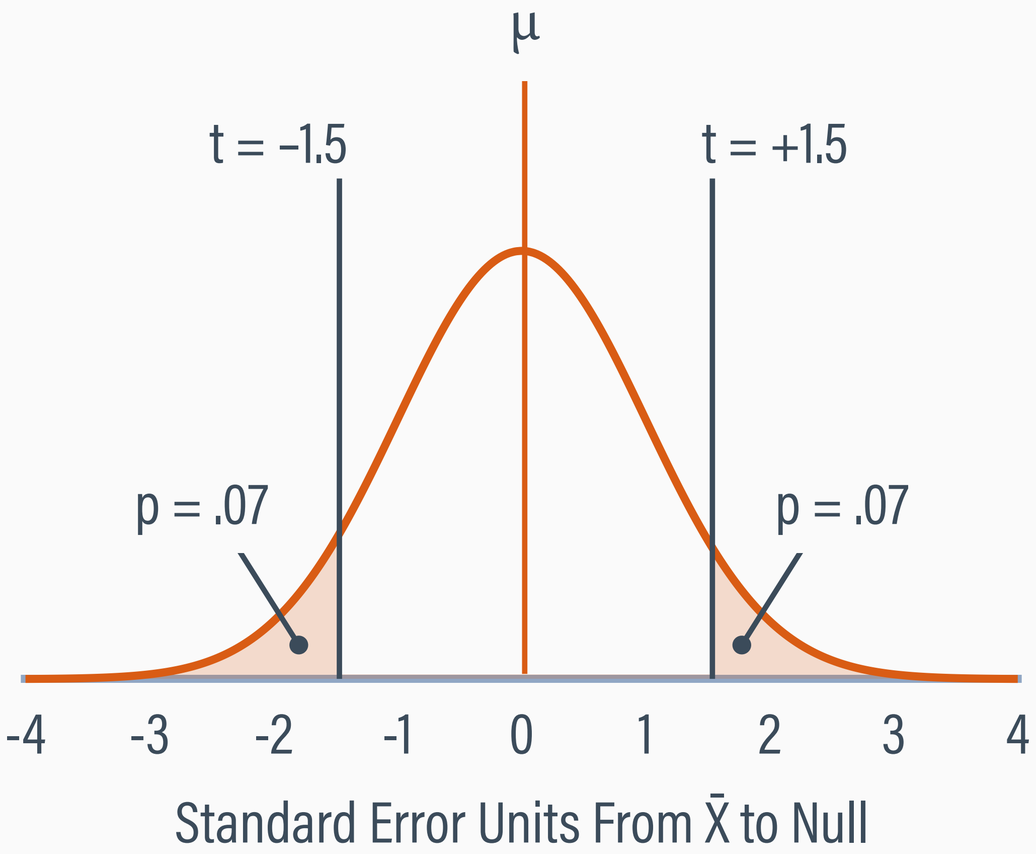
- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ±1 is .32 (.16 positive + .16 negative)



P-VALUE AT ± 1.5 STANDARD ERRORS

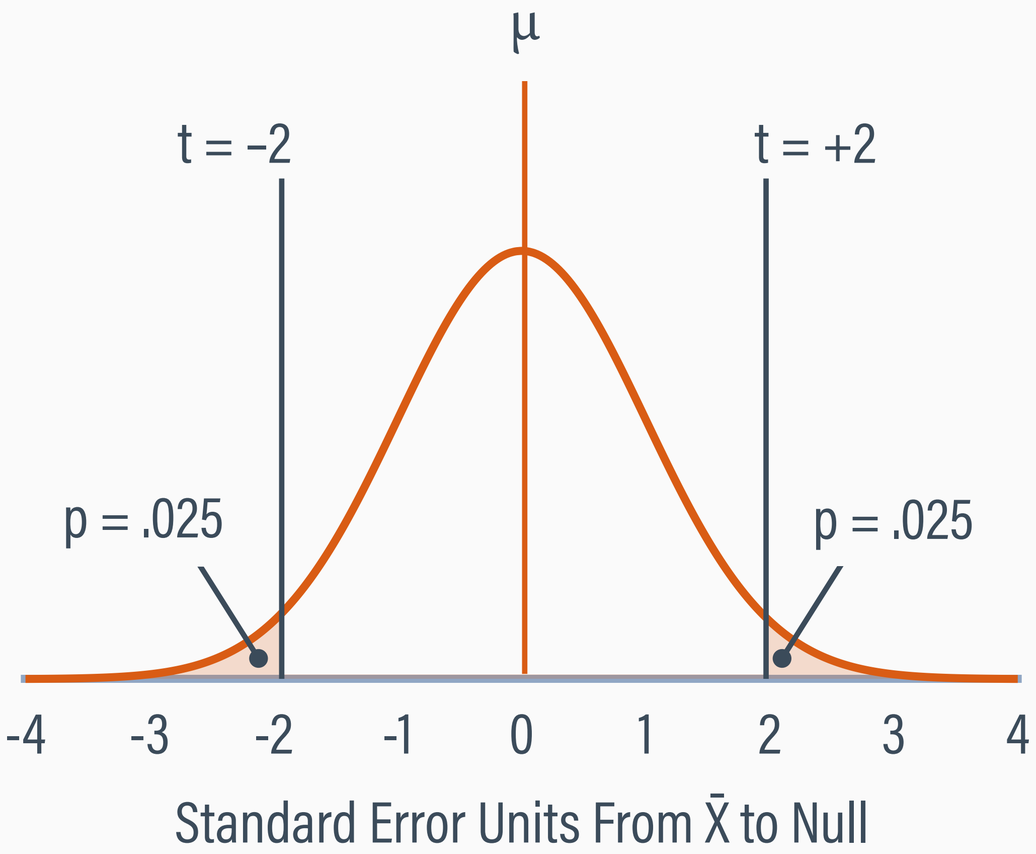
- 7% of all hypothetical samples are +1.5 or more standard error units above the null mean, and 7% are –1.5 or more standard errors below

- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ±1 is .14 (.07 positive + .07 negative)



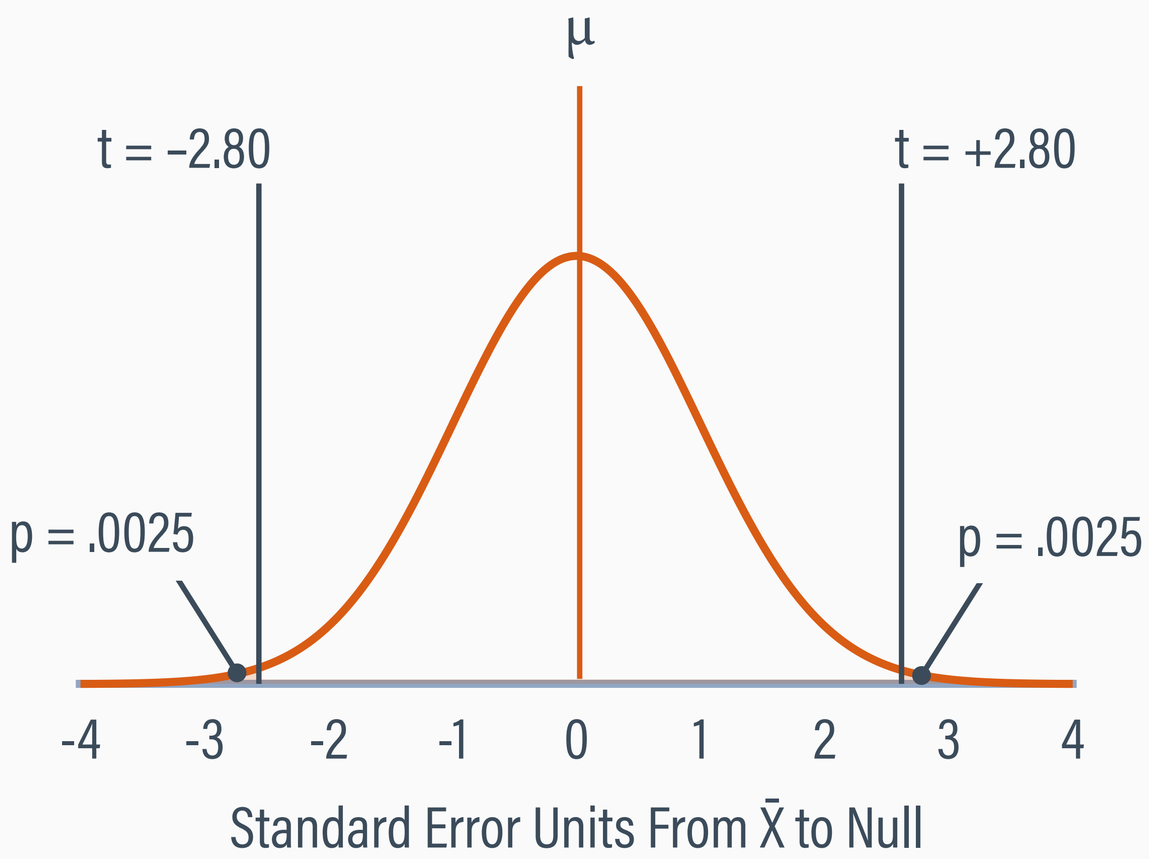
P-VALUE AT ± 2 STANDARD ERRORS

- 2.5% of all hypothetical samples are at least +1.96 standard error units above the null mean, and 2.5% are –1.96 or more below
- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ±1.96 is .05 (.025 positive + .025 negative)



TWO-TAILED P-VALUE FOR DEPRESSION DATA

- Depress mean: $\bar{X} = 14.85$, $t = 2.80$
- The two-tailed probability of drawing a sample from the null population with a t-statistic of at least ± 2.80 is $p = .005$
- The probability of drawing a sample mean at least as extreme as ours from a null population with $\mu = 13$ is about 0.50%

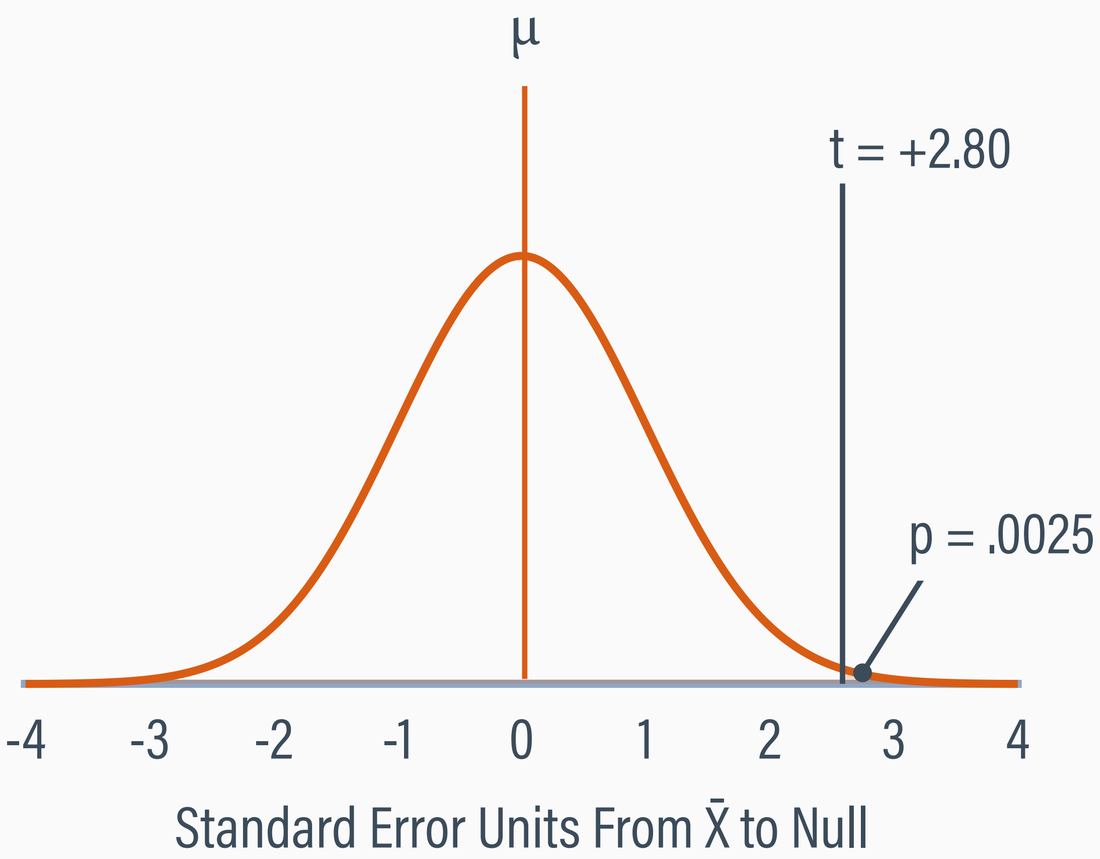




Consider the following statement, which is a common misinterpretation about the probability value: p = .005 means there's a 0.50% chance the null is true (and 99.5% chance the alternative is true). In small groups of two or three, discuss why this interpretation is incorrect. Hint: Think about the frequentist statistical paradigm.

ONE-TAILED P-VALUE FOR DEPRESSION DATA

- Depress mean: $\bar{X} = 18.10$, $t = 2.80$
- Suppose the researchers specified a one-tailed hypothesis where only an increase in depression could refute the null
- The one-tailed probability of drawing a sample from the null population with a t-statistic of at least +2.80 is $p = .0025$



SIGNIFICANCE TESTING STEPS

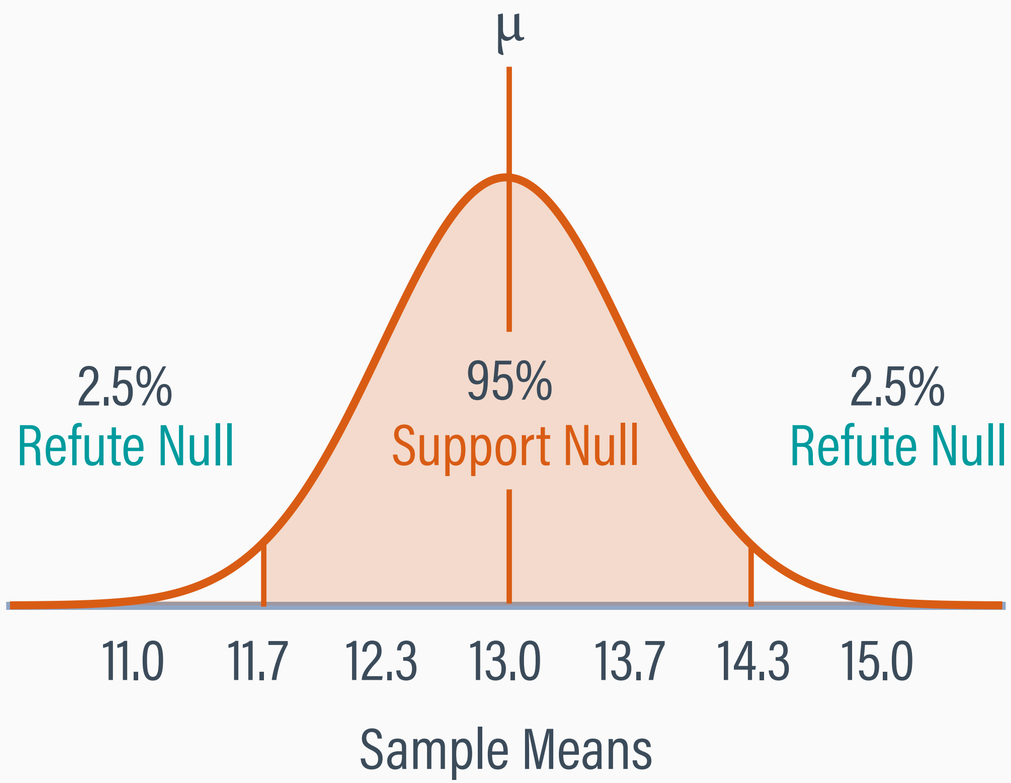
1. Specify hypotheses about population
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4. Compare data to null hypothesis
5. Evaluate hypotheses and draw conclusion

RESEARCH QUESTION REVISITED

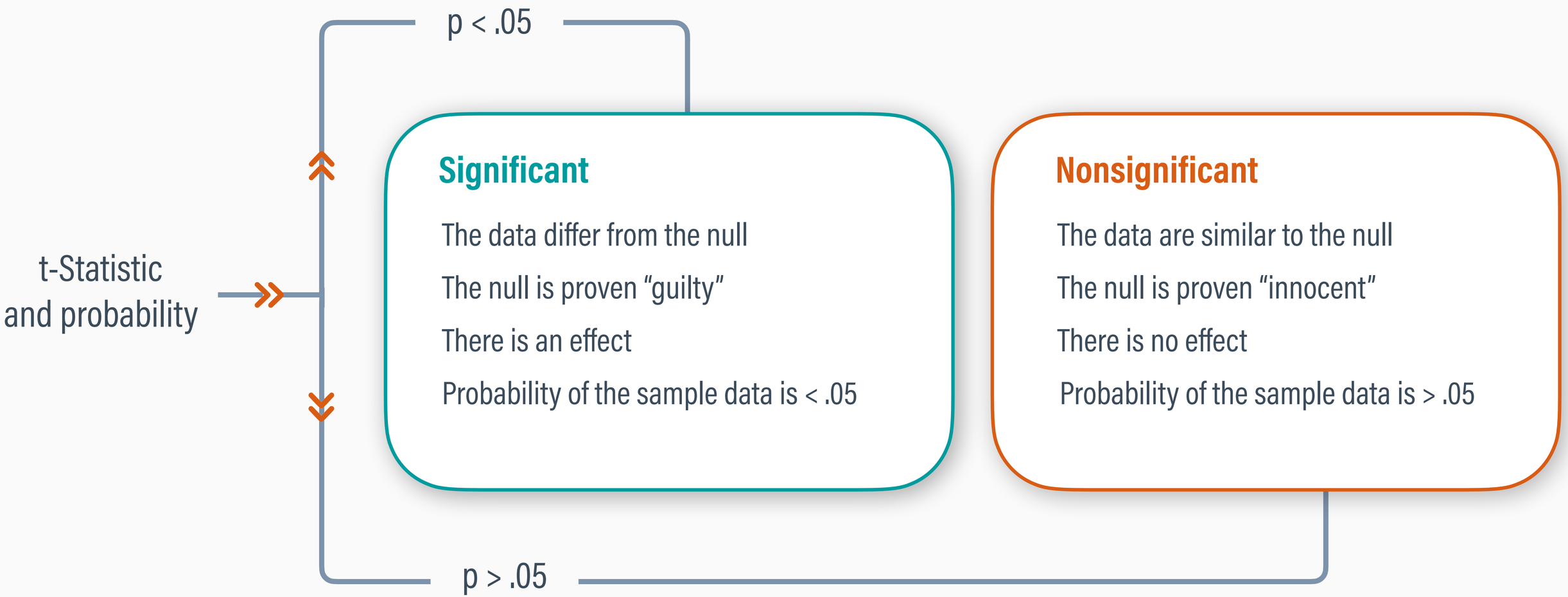
- Studies typically attempt to answer a handful of research questions involving associations between key variables
- Do adults undergoing diagnostic evaluation for a possible eye cancer experience depressive symptoms at the level typical of cancer patients awaiting treatment?
- CES-D scores ~ 13 are normative for cancer patients

5% SIGNIFICANCE CRITERION REVISITED

- By convention, we refute the null if the sample $\bar{X}$ falls outside the middle 95% of the sampling distribution ($p < .05$)
- Such as sample has less than a 5% chance of originating from the null population
- We deem the null implausible because our data are unlikely to originate from that population

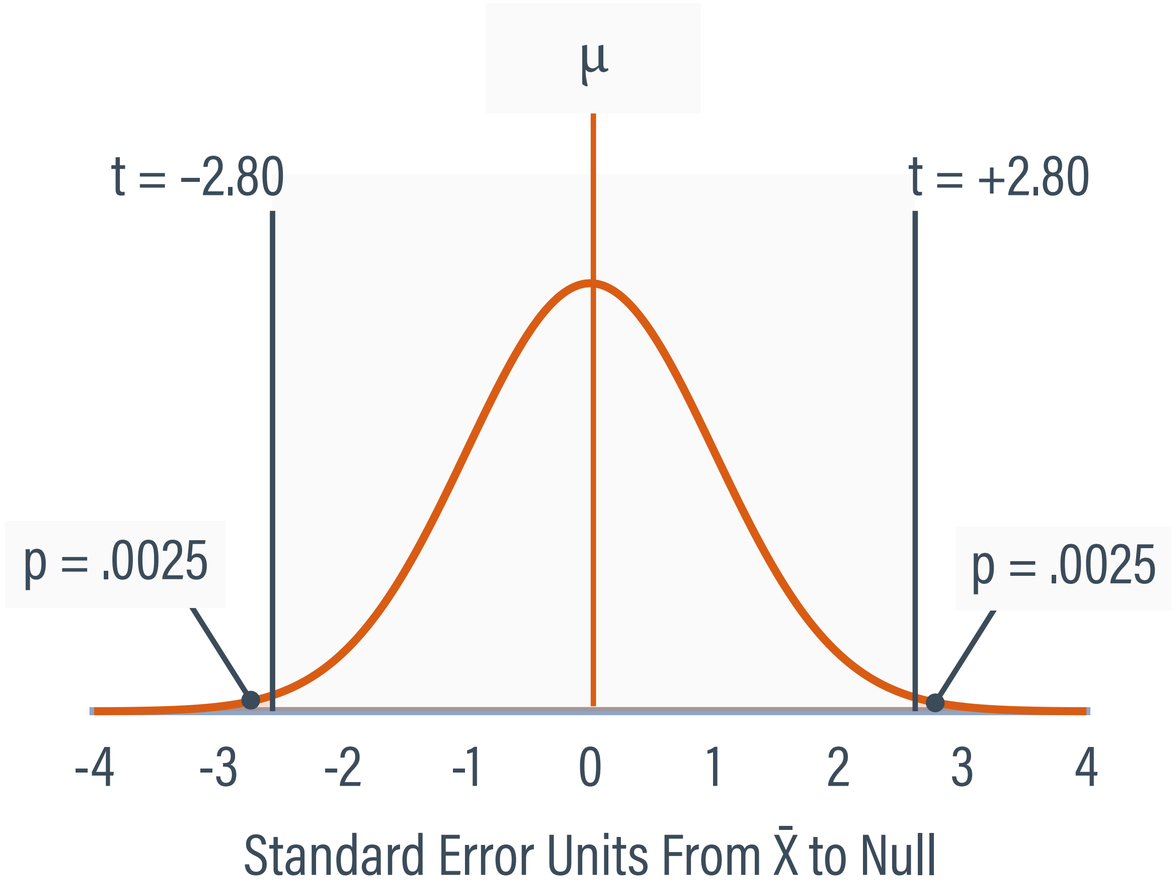


DECISION TREE





The two-tailed probability for the depression study is p = .005. In small groups of two or three, discuss your decision about the the null hypothesis. Translate your decision into a tangible statement about the null.



CONCLUSION: TWO-TAILED ALTERNATE

- The p-value of .005 (0.50%) would lead us to refute the null
- A mean as large as $\bar{X} = 14.85$ is unlikely to have plausibly originated from a null population with $\mu = 13$
- Adults evaluated for a possible eye cancer differ from the norm in cancer patients; the data suggest they have higher depression

OUTLINE

1. Quick review
2. Overview of NHST
3. Significance testing steps
4. Study questions

STUDY QUESTIONS

- The study questions continue using the ocular cancer research scenario introduced in the lecture
- The questions illustrate the kinds of statistical reasoning the exams will require, including interpreting evidence, evaluating conclusions, and connecting graphical and numerical results
- Midterm exam questions will require you to apply these same reasoning skills to new research scenarios

STUDY QUESTIONS

Use the following research scenario to answer the study questions:

Researchers want to know whether adults evaluated for a possible eye cancer differ in dispositional optimism from adults in the general population. Optimism was measured with the three-item optimism subscale of the Life Orientation Test–Revised (LOT-R), which asks people to rate their agreement with statements such as "In uncertain times, I usually expect the best." Scores range from 0 to 12, with higher scores indicating greater optimism. In the sample of $N = 299$ adults, the researchers obtain a sample mean of $\bar{X} = 8.56$. A large population-based study reported a mean of 9 on the optimism subscale in the general adult population. To evaluate optimism in this sample, you will perform significance testing steps assuming a population with a true mean of $\mu = 9$ (i.e., a population where adults evaluated for a possible eye cancer are no different in optimism from adults in general).

STUDY QUESTIONS (1)

1. State the null hypothesis, both as a sentence and using statistical symbols.

2. State the two-tailed alternate hypothesis, both as a sentence and using statistical symbols.

STUDY QUESTIONS (2)

3. State a one-tailed alternate hypothesis of your choosing (devise one that you think makes sense), both as a sentence and using statistical symbols.

4. Explain whether you think a one- or two-tailed alternate hypothesis is most appropriate for this research scenario.

R OUTPUT

```
   One Sample t-test

data:  Cancer$Optimism
t = -2.572, df = 298, p-value = 0.0106
alternative hypothesis: true mean is not equal to 9

95 percent confidence interval:
 8.226635 8.897111

sample estimates:
mean of x
 8.561873
```

STUDY QUESTIONS (3)

5. The sampling distribution under the null hypothesis plays a vital role in hypothesis testing. Explain how the 5% significance criterion is applied to this distribution, and how it is used to decide whether to reject the null hypothesis.

6. A sample of $N = 165$ participants had a mean of $\bar{X} = 8.56$. The test statistic was $t = -2.57$. Explain what the t-statistic measures. What do the sign and the magnitude of the t-statistic indicate about the plausibility of the null hypothesis?

STUDY QUESTIONS (4)

7. Using a two-tailed test, the researchers report the results as “statistically significant.” What does this imply about whether a sample mean of $\bar{X} = 8.56$ is likely to have come from a null population with $\mu = 9$?

8. Suppose instead that they reported the results as “statistically non-significant.” What does this imply about whether a sample mean of $\bar{X} = 8.56$ is likely to have come from a null population with $\mu = 9$?

STUDY QUESTIONS (5)

9. The two-tailed p-value was .01. Provide an interpretation of the probability value.

10. Consider the following statement, which is a common misinterpretation about the probability value: p = .01 means there's a 1% chance the null is true (and 99% chance the alternative is true). Discuss why this interpretation is incorrect.

11. Still referring to the p-value of .01, what does this tell you about whether patients screened for cancer differ from the general population?